

Analysis of Algorithm

Q1: what is the time complexity of the given function?

```
def unique1(S):
    for j in range(len(S)):
        for k in range(j+1, len(S)):
            if S[j] == S[k]:
                return False # found duplicate pair
    return True
```

- A. $O(n^3)$
- B. $O(n)$
- C. $O(\log n)$
- D. $O(n^2)$

Solution: D

////////////////////////////////////

Q2: what is the time complexity of the given function?

```
def ex2( n ):
    count = 0
    for i in range( n ) :
        count += 1
    for j in range( n ) :
        count += 1
    return c
```

- A. $O(n^3)$
- B. $O(n)$
- C. $O(\log n)$
- D. $O(n^2)$

Solution: B

////////////////////////////////////

Q3: what is the time complexity of the given function?

```
for i in range (1, n + 1):  
    for j in range (1, m + 1):  
        x = x + 1  
    for k in range (1, p + 1):  
        x = x + 1
```

- A. $O(n)$
- B. $O(n(m+p))$
- C. $O(m+p)$
- D. $O(m*p)$

Solution: B

////////////////////////////////////

Q4: What is the dominant term of the function $f(n) = 100n \cdot \log(n) + n^3 + 100n$ in O notation?

- A. $O(n^3)$
- B. $O(n)$
- C. $O(\log n)$
- D. $O(n^2)$

Solution: A

Arrays and Lists

Q5: you have an array Arr_array with 10 elements with the data value from 1 to 10. What will be the value of the array Arr_array after executing this code:

```
for i in range(0, 9):  
    a[i + 1] = a[i]
```

- A. Array Values will be from 1 to 10
- B. All Array values will be 1 except the first element
- C. All Array values will be 1 except the last element
- D. All Array values will be 1

Solution: D

```
In [16]: a = [1,2,3,4,5,6,7,8,9,10]  
         for i in range(0,9):  
             a[i+1] = a[i]  
  
         print(a)  
  
[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
```

////////////////////////////////////

Stack and Queue

Q6: what we will have in the stack after running this code?

```
values = Stack()
for i in range(16):
    if i % 3 == 0:
        values.push(i)
    elif i % 4 == 0:
        values.pop()
```

- A. 0,3,6,9,12,15
- B. 0,6,9,12,15
- C. 0,9,12,15
- D. 0,9,15

Solution: C

Stack can be considered as a list. So:

```
values = []
for i in range(16):
    if i%3 == 0:
        values.append(i)
    elif i%4 == 0:
        values.pop()

print(values)
```

[0, 9, 12, 15]

////////////////////////////////////

Q7: what we will have in the Queue after running this code?

```
values = Queue()
for i in range(16):
    if i % 3 == 0:
        values.enqueue(i)
```

- A. 0,3,6,9,12,15

- B. 0,6,9,12,15
- C. 0,9,12,15
- D. 0,9,15

Solution: A

////////////////////////////////////

Linked list

Q8: **Linked list** is considered as an example of _____ type of memory allocation.

- A. Dynamic
- B. Static
- C. Compile time
- D. Heap

Solution: A

////////////////////////////////////

Q9: given function is going to find the sum of the nodes values in the linked-list. Fill in the blank.

Def ABC (head):

P = head

Sum = head.data

While p.next != head:

Return sum

- A. Sum += p.data, p= p.next
- B. Sum = p.data, p= head. Next
- C. head= p.next, Sum += p.data
- D. p= head. Next, Sum = p.next.data

Solution: A

////////////////////////////////////

Q10: what this function will do in the circular linked list?

```
p = head
while p.next.next != head:
    p = p.next

p.next.next = head.next
head.next = p.next
p.next = head
head = head.next
return head
```

- A. Calculates the number of nodes in the linked list
- B. Changes the head of the linked list
- C. Changes a single linked list to a circular linked list
- D. Exchange first and last nodes using head and p

Solution: D

////////////////////////////////////

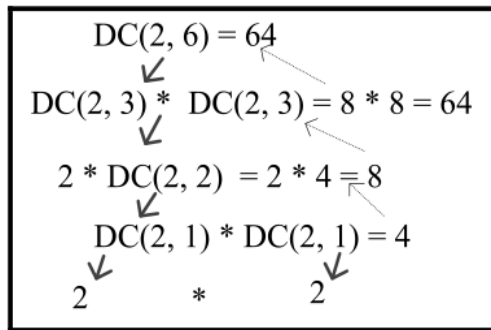
Recursion

Q11: in the given recursive function, what would be the final result of function with the given inputs a = 2 and n = 6

```
def Dc(a, n):
    if n == 1:
        return a
    if n % 2 == 0 :
        return(DC(a, n // 2) * DC(a, n//2))
    return (a * DC(a, n - 1))
```

- A. 4
- B. 8
- C. 64
- D. 32

Solution: C



////////////////////////////////////

Sorting algorithm

Q12. If the given input array is sorted or nearly sorted, which of the following algorithm gives the best performance?

- A. Insertion sort
- B. Selection sort
- C. Quick sort
- D. Merge sort

Solution: A

////////////////////////////////////

Q13: Trace the **Insertion algorithm** over the array here. A = [12, 11, 13, 5, 6]. Take the right most element as the pivot.

What would be the array after 3 pass of sorting?

- A. [11, 12, 13, 5, 6]
- B. [12, 11, 13, 5, 6]
- C. [11, 5, 12, 13, 6]
- D. [5, 11, 12, 13, 6]

Solution: D

12,11,13,5,6

1: 12|11,13,5,6 >>> 11,12,13,5,6

2: 11,12|13,5,6 >>> 11,12,13,5,6

3: 11,12,13|5,6 >>> 5,11,12,13,6

////////////////////////////////////

Hashing

Q14: The collision resolution strategy of looking at the next bucket to see if it is available is which of the following?

- A. Linear probing
- B. Quadratic probing
- C. Double hashing
- D. None of the above

Solution: A

////////////////////

Q15: A linear-probing hash table of length 10 uses the hash function $h(x) = x \bmod 10$. After inserting six integer keys into an initially empty hash table, the array of keys is:

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

Which of the following choice(s) are insertion sequences resulting in the above hash table? Assume that the length of the hash table does not change during the insertions. Check all that apply.

- A. 46, 42, 34, 52, 23, 33
- B. 34, 42, 23, 52, 33, 46
- C. 46, 34, 42, 23, 52, 33
- D. 42, 23, 34, 52, 46, 33

Solution: it could be C or D (each one has been selected will be graded as correct)

////////////////////

Greedy Algorithm

Q16: Some famous algorithms which each is based on a famous approach are given here. which of the following is correct?

- 1. Prims algorithm**
- 2. Dijkstra shortest path algorithm**
- 3. Kruskal minimum spanning tree algorithm**
- 4. Merge sort algorithm**
- 5. Bellman-ford algorithm**
- 6. Quick sort**

- a) Divide and Conquer**
- b) Greedy**
- c) Dynamic Programming**

- A. 1(c), 2(c), 3(b), 4(a), 5(c), 6(a)
- B. 1(b), 2(b), 3(b), 4(a), 5(c), 6(a)
- C. 1(b), 2(b), 3(c), 4(a), 5(b), 6(a)

Solution: B

Greedy: Prims algorithm, Dijkstra shortest path algorithm, Kruskal minimum spanning tree algorithm

Divide and Conquer: Merge sort algorithm, Quick sort

Dynamic Programming: Bellman-ford

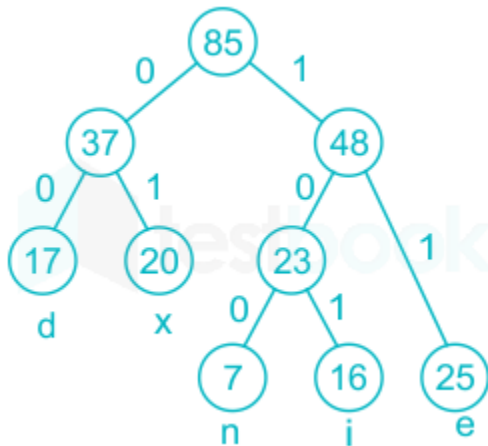
////////////////////////////////////

Huffman Coding

Q17: A text is made up of characters from set of letters $S = \{I, N, X, E, D\}$. The frequency of letters are 16, 7, 20, 25, 17 respectively. Which of the following code is the Huffman coding of letter N?

- A. 100
- B. 11
- C. 01
- D. 00

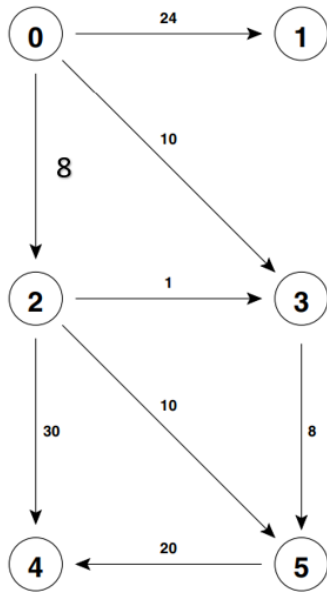
Solution: A



////////////////////////////////////

Graph

Q18: using Dijkstra algorithm, what would be the path cost from node 0 to the other nodes?



- A. [24,8,30,18,38]
- B. [24,8,9,17,37]
- C. [24,8,30,17,38]
- D. [24,8,9,18,38]

Solution: B

////////////////////////////////////

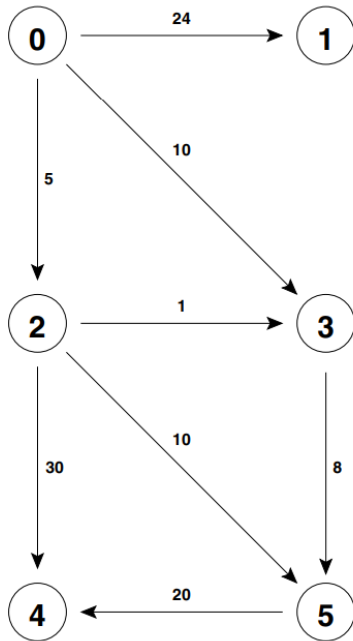
Q19: A person wants to visit some places. He starts from a vertex and then wants to visit every vertex till it finishes from one vertex, backtracks and then explore other vertex from same vertex. What algorithm he should use?

- A. Breadth First Search
- B. Prim's algorithm
- C. Depth First Search
- D. Kruskal's Algorithm

Solution: C

////////////////////////////////////

Q20: Starting from node 0. What would be a valid Depth first search (DFS) sequence?



- A. 0,1,2,3,5,4
- B. 0,1,2,3,4,5
- C. 0,2,3,1,5,4
- D. 0,2,3,1,4,5

Solution:

A

////////////////////////////////////

Q21: Depth First Search and Breath First Search algorithms uses _____ and _____ data structure for its implementation respectively.

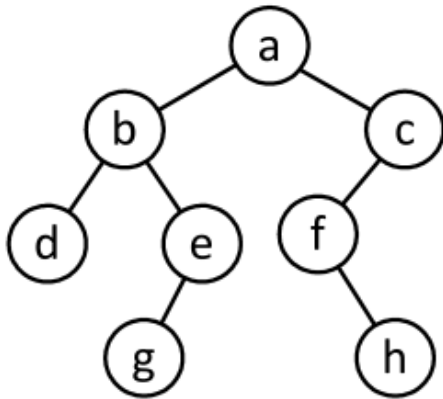
- A. Queue, Stack
- B. Stack, Queue,
- C. Stack, priority Queue
- D. Priority Queue, Stack

Solution: B

////////////////////////////////////

Tree

Q22: which of the following is a valid pre-order sequence?



- A. abcdefgh
- B. abgeafhc
- C. abdegcfh
- D. dgebhfca

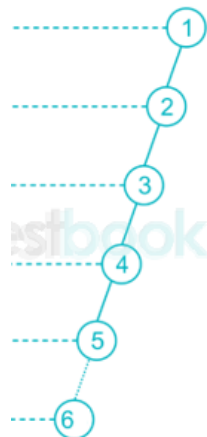
solution: C

////////////////////////////////////

Q23: the maximum height of a binary tree with N nodes is _____?

- A. N
- B. N-1
- C. N+1
- D. N+N

Solution: B



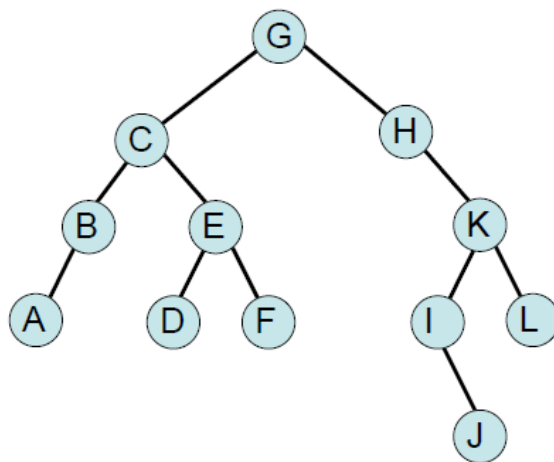
We have 6 nodes, but as you see the height is 5. Because there are 5 edges from root to the deepest leaf node.

Height of a Node : The height of a node is the number of edges from the node to the deepest leaf (ie. the longest path from the node to a leaf node).

Height of a Tree : The height of a Tree is the height of the root node or the depth of the deepest node

////////////////////////////////////

Q24: A post-order traversal of the following tree visits the nodes in which order?



- (a) G C B A E D F H K I J L
- (b) A B C D E F G H I J K L
- (c) A B D F E C J I L K H G
- (d) All of these are valid post-order traversals.

Solution: C

////////////////////////////////////

Q25: find the 2nd key in the preorder traversal of a Binary Search Tree (BST) with the given postorder?

Postorder: 5,6,15,10,23,24,22,26,20

- A. 26
- B. 10
- C. 22
- D. 24

Solution: B

Preorder: 20,10,6,5,15,26,22,24,23

